# RecoverAI: Predictive Deleted File Recovery with Semantic Search

## CARP: Confidence-Adaptive Recovery Prioritization

**Universal Project Master Defense Handbook & Comprehensive External Viva Guide**

**Institution:** VIT AP UNIVERSITY
**Department:** SCOPE (Computer Science & Engg.)
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**CORE EXAMINER TALKING POINT:** *"RecoverAI solves the blind brute-force file carving dilemma in digital forensics. Traditional tools scan unallocated space sector-by-sector with zero knowledge of block integrity. RecoverAI introduces CARP (Confidence-Adaptive Recovery Prioritization): an ML-driven framework that predicts file recoverability BEFORE carving, optimally budgets investigator time via a 0/1 knapsack allocator, and makes all recovered artifacts searchable via dense semantic embeddings."*

## 1. Project Team Roster & Module Defense Ownership

Every team member must be ready to explain the complete project end-to-end, defend their primary module in technical depth, and seamlessly answer cross-cutting questions using the following collaborative ownership matrix:

| Team Member | Assigned Viva Role | Primary Modules & Architecture | Key Defense Responsibility |
|---|---|---|---|
| **Abdul Rahamtulla** abdul9866 | **Adversarial Verification** | PyTorch MLP, Soft-Voting Ensemble, Knapsack 0/1 Allocator, Adversarial corruptions | Defense of MLP layers, BCE/MSE loss, ensemble soft voting, knapsack budget allocation, and SHA-256 verdict integrity. |
| **Asmin Shaik** Asminshaik25 | **Research & Threat Model** | Synthetic Corpus (15 files), 24 Deletion Conditions, xCarver v4 file carver, Forensic threat matrix | Defense of ground truth generation, SSD TRIM mechanics, sector-level fragmentation, raw carving vs metadata carving. |
| **Polusani Shashwath** P-SHASHWATH | **Dependency & Recovery Surface** | Docker containerization, SQLite relational schema (7 tables), OS environment isolation, Pipeline QA | Defense of database schema integrity, reproducible builds, ACID transactions, memory footprint, and production deployment. |
| **Koduru Venkata Chandrika** koduruvenkatachandrika2006-cell | **Evaluation & Reporting** | ChromaDB Vector Store, SentenceTransformer (all-MiniLM-L6-v2), Fused Search (Dense+BM25), Metrics | Defense of 100% P@1 search, cosine similarity vs dense embeddings, 5-fold cross-validation results, and analytical reporting. |

## 2. Executive Problem Statement & Motivation

**The Forensic Dilemma:** In modern digital investigations, examiners are confronted with storage drives scaling into multiple terabytes. Standard carving tools (e.g., Scalpel, Foremost, PhotoRec) execute exhaustive byte-by-byte scans across unallocated sectors. However:
• **SSD TRIM Commands:** Modern solid-state drives actively zero out unallocated Logical Block Addresses (LBAs) via background garbage collection, causing traditional carvers to waste hours reading contiguous blocks of pure zeros (`0x00`).
• **Wear-Leveling & Fragmentation:** Storage controllers scatter files non-contiguously, rendering standard header-footer reassembly invalid.
• **Overwrites & Active Data:** Partial overwrites destroy internal headers, producing corrupt, unopenable recovered files.
• **Unorganized Outputs:** Carvers dump thousands of raw files named `carved_000124.pdf`, requiring manual inspection with no semantic context.
**RecoverAI Solution:** Pre-carving recoverability prediction ($R \in [0, 1]$) via a soft-voting ensemble (82.70% accuracy, 83.71% AUC), knapsack-constrained allocation reducing False Recovery Rate (FRR) to 15.18%, and dense semantic retrieval (100% P@1).

## 3. End-to-End System Architecture & Data Pipeline

The RecoverAI architecture is structured into four sequential, tightly coupled layers:

```
[Layer 1: Corpus & Deletion Engine] 15 base files → 24 forensic deletion conditions → Raw disk image with injected corruption.
[Layer 2: Extraction & Verdict DB] xCarver v4 extracts sectors → 12 metadata/content features → SQLite DB (7 relational tables).
[Layer 3: Predictive ML & Knapsack] XGBoost + PyTorch MLP → Soft-Voting Ensemble ($R$) → Dynamic Programming 0/1 Knapsack.
[Layer 4: Artifact Index & Semantic Search] Recovered text extraction → `all-MiniLM-L6-v2` 384-d vectors → ChromaDB + BM25 Fusion.
```

## 4. Experimental Corpus, Deletion Grid & Ground Truth

**Synthetic Corpus:** 15 representative files across 7 critical forensic file formats: PDF, DOCX, JPG, PNG, MP3, MP4, and ZIP. This diverse format distribution ensures cross-format evaluation covering structured text, compressed containers, raster graphics, and multimedia.

**The 24 Deletion Conditions (4 Threat Tiers):**
1. **Tier 1 (Metadata Unlinking):** Master File Table (MFT) / inode record marked free, but data sectors remain 100% intact.
2. **Tier 2 (TRIM Emulation):** Deterministic zeroing (`0x00`) applied across unallocated LBA ranges, simulating SATA/NVMe Deallocate commands.
3. **Tier 3 (Sector Fragmentation):** Clusters scattered non-contiguously across disk tracks with interlaced slack noise.
4. **Tier 4 (Progressive Overwrites):** Partial overwriting of 25%, 50%, 75%, and 100% of file sectors with pseudorandom bytes.

**Ground Truth Verification Engine:** Total of 1,260 experiment records. The system computes the SHA-256 hash of the pre-deletion file and compares it against the carved artifact: Exact Match ($R = 1.0$), Corrupt Partial Carve ($0 < R < 1.0$), or Completely Destroyed ($R = 0.0$).

## 5. Pre-Carving Feature Engineering (12 Core Features)

Before spending computational budget carving blocks, RecoverAI extracts 12 low-overhead features from filesystem metadata and sector headers:

• file_size_bytes: Original recorded file size in bytes.
• time_since_deletion_sec: Elapsed time between deletion timestamp and acquisition.
• is_contiguous: Boolean flag (1 if contiguous cluster chain, 0 if fragmented).
• fragment_count: Number of fragmented cluster segments.
• trim_enabled: Boolean indicator of whether TRIM/Deallocate was triggered.
• sector_entropy: Shannon entropy of the first 512-byte sector ($H = -\sum p_i \log_2 p_i$; 0.0 = zeroed, 7.8+ = encrypted/compressed).
• magic_bytes_present: Signature match verification for format header (e.g., `%PDF`, `PK\x03\x04`, `\xFF\xD8\xFF`).
• mft_record_intact: Structural integrity of the filesystem metadata descriptor.
• cluster_overlap_ratio: Percentage of clusters reassigned to active files.
• storage_media_type: HDD (magnetic) vs. SSD (flash with garbage collection).
• file_type_cat: Categorical encoding of MIME format.
• allocation_gap_blocks: Physical sector distance between non-contiguous allocated extents.

## 6. Machine Learning Model Architecture & Validation

RecoverAI employs a soft-voting ensemble combining a gradient-boosted decision tree baseline with a deep Multi-Layer Perceptron (MLP):
• **XGBoost Regressor / Classifier:** 100 gradient-boosted decision trees, max depth = 5, learning rate = 0.05. Captures non-linear feature interactions and threshold cutoffs across tabular forensic descriptors.
• **PyTorch Deep MLP:** Fully connected neural network architecture: Input(12) → Linear(64) → BatchNorm1d → ReLU → Dropout(0.20) → Linear(32) → ReLU → Linear(1) → Sigmoid. Trained using Adam optimizer (lr = 0.001, weight decay = $10^{-4}$) under combined Binary Cross-Entropy (classification) and Mean Squared Error (regression) loss functions.
• **Soft-Voting CARP Ensemble:** Fuses normalized probability outputs: P_ens = 0.50 × P_xgb + 0.50 × P_mlp. Ensembling dampens model variance, smoothing boundary predictions and outperforming both individual models across every benchmark metric.

| Model / Framework | Architecture / Type | Accuracy | F1 (Weighted) | AUC-ROC | MAE | $R^2$ Score |
|---|---|---|---|---|---|---|
| **XGBoost (Baseline)** | Gradient Boosted Trees (100 est, depth 5) | 81.59% | 0.7904 | 0.8193 | 0.1713 | 0.1779 |
| **PyTorch MLP (Main)** | Dense NN (64-32-1) + BatchNorm + Dropout | 82.06% | 0.7750 | 0.8360 | 0.1739 | 0.2786 |
| **Soft-Voting Ensemble ★ RecoverAI Best ★** | Confidence Average (0.50 XGB + 0.50 MLP) | **82.70%** | **0.7915** | **0.8371** | **0.1692** | **0.2781** |

**Rigorous Evaluation Protocol:** Evaluated under **5-Fold Stratified Out-of-Fold (OOF) Cross-Validation**. No data leakage between training and validation folds. AUC-ROC is the primary forensic metric (0.8371) because forensic triage depends on accurate relative ranking of evidence rather than an arbitrary 0.5 classification threshold.

## 7. Knapsack Resource Optimizer (Forensic Prioritization)

**The Mathematical Problem:** In incident response, examiners have a fixed time budget $B$ (e.g., 0.5s per deletion condition, or 30 minutes for a drive). Exhaustive carving of every candidate sector exceeds $B$ by $10\times$ to $50\times$. RecoverAI models recovery as a **0/1 Knapsack Optimization Problem**:

$$\max \sum_{i=1}^{N} u_i \cdot x_i \quad \text{subject to} \quad \sum_{i=1}^{N} c_i \cdot x_i \le B, \quad x_i \in \{0, 1\}$$

where $u_i = R_i \times \text{EvidencePriority}_i$ is the expected forensic utility, $c_i$ is the computational carving cost in seconds, and $x_i$ is the binary decision variable. Based on predicted recoverability $R$, RecoverAI selects one of four dynamic effort strategies:

| Strategy | Trigger Condition | Execution Action | Time Cost | Forensic Utility |
|---|---|---|---|---|
| **skip** | Predicted R < 0.20 or TRIM zeroed | Bypass cluster parsing entirely; record unrecoverable verdict | 0.001 s | 0.00 (Zero effort spent) |
| **header_check** | 0.20 ≤ Predicted R < 0.45 | Read first 512 bytes; verify file signature / magic numbers | 0.050 s | 0.20 × R (Early filter) |
| **scoped** | 0.45 ≤ Predicted R < 0.70 | Extract contiguous cluster blocks up to estimated header length | 0.200 s | 0.60 × R (Partial recovery) |
| **full** | Predicted R ≥ 0.70 (High confidence) | Exhaustive multi-cluster carving, sector stitching & SHA validation | 0.500 s | 1.00 × R (Maximum evidence) |

**Forensic Impact:** Knapsack optimization reduces the False Recovery Rate (FRR) to **15.18%**. This eliminates 84.8% of wasted carving cycles on corrupted or TRIM-zeroed sectors, allowing the examiner to recover 91.4% of total salvageable data within a fraction of the time.

## 8. Semantic Search & Vector Retrieval Engine

**The Need for Semantic Search:** File carving outputs unorganized, numerically indexed files (`carved_00042.pdf`). Examiners do not have filenames or folder hierarchies. Keyword search fails when documents use synonyms, paraphrasing, or OCR transcription errors.
**Embedding Architecture:** RecoverAI uses the all-MiniLM-L6-v2 SentenceTransformer model to map extracted text into **384-dimensional dense semantic vector space**. Text is chunked (512 tokens with 50-token overlap) and indexed into a local **ChromaDB** vector store.
**Hybrid Fused Search (Dense Vector + BM25):** To ensure both exact keyword hits and high-level conceptual discovery, RecoverAI computes a fused retrieval score:
$$S_{fused}(q, d) = \alpha \cdot S_{vector}(q, d) + (1 - \alpha) \cdot S_{BM25}(q, d), \quad \alpha = 0.70$$
**Search Benchmarks:** Achieves **Precision@1 = 100% (1.000)** and **Mean Reciprocal Rank (MRR) = 1.000** across all domain test queries (e.g., query *"James Webb telescope exoplanet"* instantly retrieves astronomy_article.txt as the top rank).

## 9. Database Schema & Storage Architecture (SQLite 7 Tables)

Designed by Polusani Shashwath, RecoverAI maintains an ACID-compliant SQLite relational database (`recoverai.db`) with 7 relational tables:

```
1. test_corpus: Tracks 15 original base files, sizes, MIME types, and pre-deletion SHA-256 hashes.
2. deletion_events: Records 24 deletion condition parameters, timestamps, target sectors, and corruption tier.
3. carving_attempts: Logs xCarver execution metrics, cluster traversal paths, elapsed CPU milliseconds, and carved file sizes.
4. hash_verdicts: Cryptographic comparison between carved output SHA-256 and pre-deletion ground truth hash.
5. extracted_features: The 12 pre-carving tabular features used by XGBoost and PyTorch MLP.
6. ml_predictions: Predicted recoverability scores $P_{xgb}$, $P_{mlp}$, and ensemble score $P_{ens}$ for each file-condition pair.
7. allocations: Knapsack effort assignments (`skip`, `header_check`, `scoped`, `full`) and budget consumption.
```

## 10. External Examiner Viva Q&A; Bank — Conceptual & Forensic Core

The following questions are rigorously compiled from typical external examiner viva interviews for digital forensics and ML projects:

**Q1: What is the main problem in traditional file carving that RecoverAI solves?**

**Answer:** Traditional file carving tools (Scalpel, Foremost, PhotoRec) are purely reactive and exhaustive—they scan unallocated sectors blindly, attempting to reassemble file headers and footers without knowing whether the underlying sectors are intact, fragmented, partially overwritten, or completely wiped by SSD TRIM. On high-capacity multi-terabyte drives, this wastes hours recovering corrupted, unopenable junk files. RecoverAI introduces CARP (Confidence-Adaptive Recovery Prioritization), predicting recoverability ($R \in [0,1]$) *before* carving, allowing optimal time-budget allocation.

**Q2: How does the SSD TRIM command affect file carving, and how does RecoverAI handle it?**

**Answer:** On modern SSDs, when an OS deletes a file, it issues an ATA TRIM or NVMe Deallocate command to the drive controller. The controller flags those physical flash pages as invalid and returns deterministic zeros (`0x00`) or random garbage upon subsequent reads during background garbage collection. Traditional carvers cannot detect this without reading every block. RecoverAI inspects sector entropy and TRIM metadata indicators: if sector entropy drops to 0.0 or a TRIM event is registered, the predicted recoverability drops to $R < 0.20$, and the knapsack engine assigns a skip strategy, saving 99.8% of carving time.

**Q3: What is the difference between metadata-based recovery and raw file carving?**

**Answer:** Metadata-based recovery relies on residual filesystem structures (such as NTFS Master File Table records or EXT4 directory entries) which still store the file's cluster runlists, file size, timestamps, and exact sector pointers. Raw file carving is performed when filesystem metadata is completely erased, formatted, or corrupted; it relies purely on content inspection—matching file signatures (magic bytes) at sector boundaries and scanning for file footers or internal structural markers.

**Q4: Why did you build xCarver v4 instead of just using PhotoRec or Scalpel?**

**Answer:** Tools like PhotoRec are standalone C binaries that execute monolithic, non-configurable carving passes. They cannot be programmatically interrupted at granular stages (such as stopping after a 512-byte header check), nor do they expose intermediate cluster traversal statistics or integrate natively with a Python ML pipeline. xCarver v4 is our purpose-built forensic carver that supports multi-mode execution (`--raw-only`, `--header-check`, `--scoped`, `--full`), logging exact timing and cluster health into SQLite for direct ML training and budget optimization.

**Q5: Explain your 24 deletion conditions. Why did you create such an extensive matrix?**

**Answer:** Real-world evidence destruction does not happen uniformly. By creating 24 distinct experimental conditions spanning 4 threat tiers (metadata unlinking, SSD TRIM zeroing, non-contiguous cluster fragmentation, and 25%, 50%, 75%, and 100% progressive overwrites), we ensured that our 1,260-sample dataset reflects the full continuum of real forensic scenarios. This prevents the ML model from merely memorizing clean binary cutoffs and trains it to accurately score partial, degraded, and completely destroyed recovery surfaces.

**Q6: How is ground truth established in RecoverAI?**

**Answer:** Ground truth is strictly established using cryptographic SHA-256 hashing. Before any deletion, the SHA-256 digest of every original file is recorded in the `test_corpus` database table. After xCarver executes under any given condition, the SHA-256 digest of the carved artifact is computed and compared against the pre-deletion hash. If the hashes match identically, ground truth $R = 1.0$. If the file is zeroed, $R = 0.0$. For partial files, a normalized similarity ratio based on valid byte sequence length is stored in the `hash_verdicts` table.

**Q7: What is sector entropy, and why is it such an important feature?**

**Answer:** Sector entropy measures the randomness of byte distribution in a 512-byte sector using Shannon entropy ($H = -\sum p_i \log_2 p_i$), ranging from 0.0 to 8.0 bits. A completely zeroed TRIM sector has $H = 0.0$. Plain text has $H \approx 3.5 - 5.0$. Compressed or encrypted formats (ZIP, JPG, AES ciphertext) exhibit high entropy ($H > 7.5$). Sector entropy serves as an instant, zero-cost discriminator to detect wiped blocks without parsing complex format grammars.

## 11. External Examiner Viva Q&A; Bank — Machine Learning & Optimization

**Q8: Why did you use an ensemble of XGBoost and PyTorch MLP instead of just one model?**

**Answer:** Tabular forensic metadata contains both sharp threshold rules (e.g., `is_contiguous == 0`, `trim_enabled == 1`) and smooth continuous interactions (e.g., `sector_entropy` interacting with `time_since_deletion_sec` and `file_size_bytes`). XGBoost excels at axis-aligned orthogonal decision boundaries, while the PyTorch MLP captures non-linear continuous manifold representations. By averaging their predicted probabilities ($0.5 \cdot P_{xgb} + 0.5 \cdot P_{mlp}$), the soft-voting ensemble reduces individual model variance and achieves superior generalization: **82.70% accuracy and 0.8371 AUC-ROC**, outperforming both individual models.

**Q9: What is the exact architecture of your PyTorch MLP?**

**Answer:** The PyTorch model is a 3-layer deep neural network designed to avoid overfitting on tabular data: Linear(12, 64) → BatchNorm1d(64) → ReLU() → Dropout(p=0.20) → Linear(64, 32) → ReLU() → Linear(32, 1) → Sigmoid(). Batch normalization stabilizes gradient flow and accelerates convergence, while 20% dropout and $L_2$ weight decay ($10^{-4}$) prevent co-adaptation of features. It is trained for 100 epochs using the Adam optimizer with a learning rate of 0.001.

**Q10: Why is AUC-ROC prioritized over raw accuracy as the primary evaluation metric?**

**Answer:** In digital forensics, classification accuracy can be deceptive due to class imbalance (e.g., when 80% of sectors are unrecoverable after a disk wipe). Furthermore, our knapsack allocator does not make decisions based on a hard 0.5 binary threshold; it sorts candidate files by their continuous predicted probabilities to maximize expected utility. AUC-ROC measures the model's ability to rank a truly recoverable file higher than an unrecoverable one across all classification thresholds, making it the mathematically rigorous metric for prioritized triage.

**Q11: Explain how you validated your models. How do you guarantee zero data leakage?**

**Answer:** We implemented **5-Fold Stratified Out-of-Fold (OOF) Cross-Validation**. In each fold, 80% of data is used for training and 20% strictly for validation. Feature scaling (StandardScaler) is fitted strictly on the training fold and applied to the validation fold to avoid data leakage. The reported 82.70% accuracy, 0.7915 F1, and 0.8371 AUC-ROC are the consolidated out-of-fold metrics across all 1,260 samples, guaranteeing that no test sample was ever seen during weight updates.

**Q12: How is the Knapsack problem formulated in RecoverAI, and why is it 0/1 rather than fractional?**

**Answer:** We formulate carving allocation as a 0/1 Knapsack problem where each file candidate $i$ has an expected utility $u_i = R_i \times \text{Priority}_i$ and a carving cost $c_i$ in seconds. The examiner sets a total forensic budget $B$ (e.g., $B = 0.5$s per condition). The objective is $\max \sum u_i x_i$ subject to $\sum c_i x_i \le B$, where $x_i \in \{0, 1\}$. It is 0/1 (discrete) because in forensic analysis, you either carve a file under a designated strategy or you skip it; carving half a cluster block without reassembly yields no usable evidential artifact.

**Q13: What are the 4 knapsack strategies and how do they save investigator time?**

**Answer:** The strategies are: skip (cost: 0.001s, for $R < 0.20$), header_check (cost: 0.05s, for $0.20 \le R < 0.45$), scoped (cost: 0.20s, for $0.45 \le R < 0.70$), and full (cost: 0.50s, for $R \ge 0.70$). If a sector is zeroed or severely corrupted, `skip` bypasses it instantly. If headers are intact, `full` recovers the complete file. This dynamic tiering reduces the False Recovery Rate (FRR) to 15.18%, cutting wasted time by 84.8%.

**Q14: What is the False Recovery Rate (FRR), and why does 15.18% matter?**

**Answer:** The False Recovery Rate is the ratio of files where full recovery effort was expended but the carved output turned out to be completely unopenable or corrupted: $\text{FRR} = \frac{\text{Failed Full Carving Attempts}}{\text{Total Carving Attempts}}$. In brute-force carvers, FRR typically exceeds 70% to 80% on wiped or fragmented disks. RecoverAI's FRR of 15.18% proves that our ML filter accurately weeds out doomed files before CPU time is spent.

## 12. External Examiner Viva Q&A; Bank — Semantic Search, Security & Systems

**Q15: Why did you choose ChromaDB and SentenceTransformers over traditional keyword search (grep/Elasticsearch)?**

**Answer:** Carved artifacts lack original filenames and metadata, and often contain OCR noise or missing headers. Traditional keyword matching (`grep` or BM25 alone) requires exact term matches; it fails if an investigator searches for *"financial fraud"* when the carved document mentions *"illicit wire transfer"* or *"hidden offshore balance"*. SentenceTransformers (`all-MiniLM-L6-v2`) embed conceptual meaning into 384-dimensional vector space, and ChromaDB provides sub-millisecond cosine similarity search, allowing investigators to find evidence by natural language meaning.

**Q16: How does Fused Search work, and why combine Dense Vectors with BM25?**

**Answer:** Pure dense vector search can occasionally miss exact forensic identifiers like serial numbers, IP addresses, or person names that have no semantic synonym. Fused Search combines the strengths of both worlds: BM25 scores lexical token matches, while vector search scores conceptual similarity. We combine them using a weighted score: $S_{fused} = 0.70 \cdot S_{vector} + 0.30 \cdot S_{BM25}$. This achieved **100% Precision@1** across all experimental query benchmarks.

**Q17: What happens if an adversary attempts anti-forensics by deliberately falsifying magic bytes?**

**Answer:** If an attacker prepends a `%PDF-1.4` magic byte header onto an executable or encrypted payload, a naive header carver is fooled. However, RecoverAI is robust against this attack because magic bytes are only 1 of 12 features. The model also inspects `sector_entropy` (which will be anomalously high for encrypted data), `file_size_bytes` consistency, and internal structural markers. Post-carving, our SHA-256 hash validation and parser integrity checks immediately flag the mismatch as an adversarial attempt, logging it in the database.

**Q18: How did you ensure database integrity and concurrency across the pipeline?**

**Answer:** Polusani Shashwath designed the SQLite database (`recoverai.db`) using strict relational foreign keys connecting all 7 tables with WAL (Write-Ahead Logging) mode enabled. WAL mode eliminates database write locks during concurrent feature extraction and model inference, ensuring ACID transactions and zero corrupted records across all 1,260 experimental trials.

**Q19: How is the system containerized, and why is Docker critical in forensic science?**

**Answer:** In digital forensics, the Daubert standard and ISO/IEC 27037 require experimental reproducibility and auditability. We packaged RecoverAI into a multi-stage Docker container encapsulating Python 3.11, PyTorch 2.2, XGBoost 2.0, ChromaDB, and xCarver v4. This eliminates host OS dependency conflicts (e.g., Windows vs. Linux filesystem drivers) and allows any external forensic lab to reproduce our exact results with a single command: docker build -t recoverai . && docker run recoverai.

**Q20: What are the main limitations of RecoverAI, and what is the future scope?**

**Answer:** Current limitations include: (1) The synthetic corpus is evaluated on 15 core files across 7 formats; scaling to proprietary CAD/database files requires expanding the format parser registry. (2) Highly fragmented files with non-sequential clusters across non-contiguous tracks still require combinatorial sector reassembly. Future work includes integrating Graph Neural Networks (GNNs) for non-contiguous cluster reassembly, hardware FPGA acceleration for wire-speed sector entropy hashing, and multi-modal search over audio and video artifacts.

## 13. Exam-Day External Guide Defense Strategy & Rapid Reference

Keep these battle-tested strategies and official numbers in mind during your oral examination:

**1. The 30-Second Elevator Pitch (Master this by heart):**
*"RecoverAI is an intelligent digital forensics framework that eliminates blind brute-force file carving. Legacy carvers waste hours scanning wiped or TRIM-zeroed blocks on modern storage. Our CARP engine predicts file recoverability before carving using a soft-voting ensemble of XGBoost and PyTorch MLP (achieving 82.70% accuracy and 0.8371 AUC), allocates forensic effort under budget constraints using a 0/1 knapsack optimizer (reducing false recovery to 15.18%), and indexes recovered evidence into ChromaDB with dense embeddings, enabling 100% P@1 natural language semantic search."*

### 2. Master Cheat-Sheet: Official Key Numbers & Metrics (Never contradict these):

| Metric / Parameter | Official Value | Forensic Significance |
|---|---|---|
| **Ensemble Accuracy** | **82.70%** | 5-Fold Out-of-Fold cross-validation across 1,260 samples. |
| **Ensemble F1-Score** | **0.7915** | Weighted F1 balancing recoverable vs destroyed classes. |
| **Ensemble AUC-ROC** | **0.8371** | Primary triage metric; evaluates rank discrimination. |
| **Knapsack FRR** | **15.18%** | False Recovery Rate reduced from >75% down to 15.18%. |
| **Search Precision@1** | **100% (1.000)** | Top-ranked document is always the correct target file. |
| **Search MRR** | **1.000** | Mean Reciprocal Rank confirms flawless top-tier retrieval. |
| **Dataset Corpus Size** | **15 files / 7 types** | PDF, DOCX, JPG, PNG, MP3, MP4, ZIP. |
| **Deletion Conditions** | **24 conditions** | Metadata, TRIM zeroing, fragmentation, 25-100% overwrite. |
| **Total Trial Records** | **1,260 records** | Stored across 7 relational SQLite tables. |
| **Embedding Model** | **all-MiniLM-L6-v2** | 384-dimensional dense semantic vectors in ChromaDB. |
| **Hybrid Fusion Weight** | **0.70 Dense / 0.30 BM25** | Optimal balance of semantic abstraction & keyword precision. |

**3. How to Handle Tricky / Curveball Questions Gracefully:**
• **If asked: "Why didn't you use a huge deep learning model like a Transformer (BERT/GPT) for recoverability prediction?"**
*Answer: "Tabular forensic metadata has only 12 features. Massive Transformer models require millions of tabular rows and are heavily prone to overfitting on low-dimensional structural features. Our PyTorch MLP with Batch Normalization and Dropout, ensembled with XGBoost, provides superior regularization, near-zero inference latency (sub-millisecond), and runs on forensic field laptops without requiring expensive multi-GPU infrastructure."*
• **If asked: "Isn't 82.70% accuracy low for digital forensics where evidence must be 100%?"**
*Answer: "RecoverAI does NOT make the final legal guilt determination—it performs triage and prioritization. In a multi-terabyte drive, an examiner cannot examine 100% of blocks. An 83.71% AUC-ROC ensures that salvageable evidence is pushed to the top of the queue. Furthermore, every file actually recovered undergoes strict cryptographic SHA-256 hash verification, so zero corrupted files are ever falsely presented as authentic evidence."*
• **If asked about a teammate's module that you don't know the exact code line for:**
*Answer: "In our architectural split, [Teammate Name] was the primary owner of [Module], while I handled [My Module]. However, from a system integration perspective, [Module] feeds [Output] into my module via [Database table/API], adhering to our strict interface specification."*

**Final Reminder:** Stay confident, speak clearly, refer to the authoritative numbers in the table above, and emphasize RecoverAI's unique end-to-end contribution from pre-carving prediction to post-carving semantic search!